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Roll no: 105

CODE import pandas

as pd

```
# Create a DataFrame data
= {
   "Name": ["Amit", "Sneha", "Rahul", "Priya"],
   "Age": [20, 21, 19, 22],
   "Marks": [85, 90, 75, 88]
}

df = pd.DataFrame(data)

print("Original Data:") print(df)

# 1. Display a column print("\nName
Column:") print(df["Name"])

# 2. Add a new column df["Grade"]
= ["A", "A+", "B", "A"]

print("\nAfter Adding Grade Column:") print(df)  # 3. Add a new row df.loc[len(df)] =
["Neha", 20, 92, "A+"]
```

```
print("\nAfter Adding New Row:")
print(df)

# 4. Update data df.loc[0,
"Marks"] = 89

print("\nAfter Updating Marks:")
print(df)

# 5. Delete a column df =
df.drop("Grade", axis=1)

print("\nAfter Deleting Grade Column:") print(df)

# 6. Filter data print("\nStudents with Marks greater
than 85:") print(df[df["Marks"] > 85])

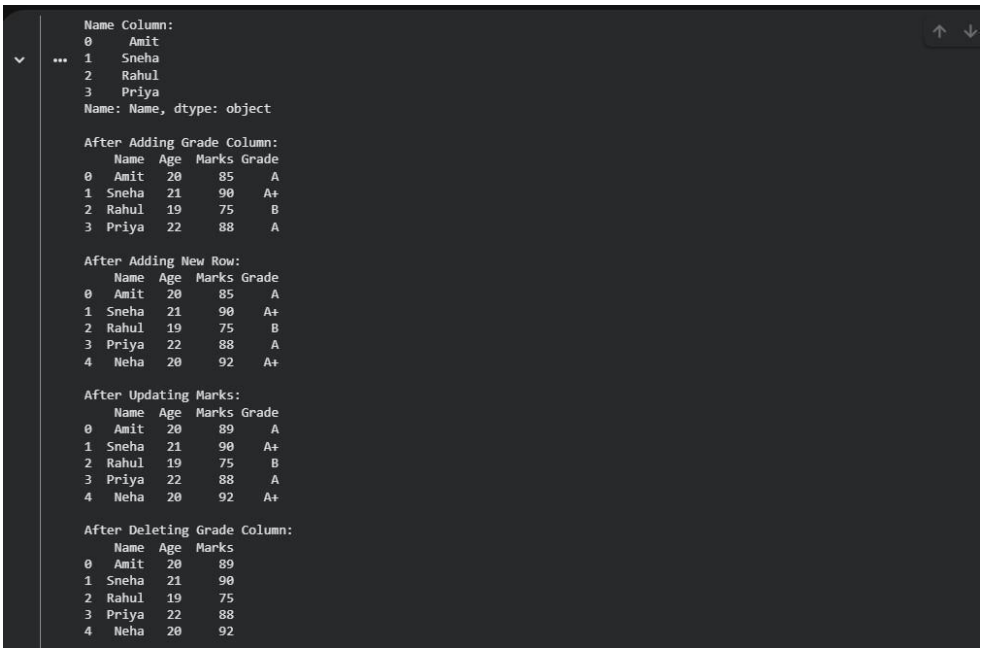
# 7. Sort data print("\nData
Sorted by Marks:")
print(df.sort_values("Marks",
ascending=False))
```

```
# 8. Calculate average
print("\nAverage Marks:")
print(df["Marks"].mean())

# 9. Find maximum marks
print("\nMaximum Marks:")
print(df["Marks"].max())

# 10. Find minimum marks
print("\nMinimum Marks:")
print(df["Marks"].min())
```

OUTPUT



```
Name Column:
0     Amit
1    Sneha
2    Rahul
3    Priya
Name: Name, dtype: object

After Adding Grade Column:
    Name  Age  Marks Grade
0   Amit   20     85     A
1  Sneha   21     90    A+
2  Rahul   19     75     B
3  Priya   22     88     A

After Adding New Row:
    Name  Age  Marks Grade
0   Amit   20     85     A
1  Sneha   21     90    A+
2  Rahul   19     75     B
3  Priya   22     88     A
4   Neha   20     92    A+

After Updating Marks:
    Name  Age  Marks Grade
0   Amit   20     89     A
1  Sneha   21     90    A+
2  Rahul   19     75     B
3  Priya   22     88     A
4   Neha   20     92    A+

After Deleting Grade Column:
    Name  Age  Marks
0   Amit   20     89
1  Sneha   21     90
2  Rahul   19     75
3  Priya   22     88
4   Neha   20     92
```

```
Students with Marks greater than 85:
    Name  Age  Marks
0   Amit   20     89
1  Sneha   21     90
3  Priya   22     88
4   Neha   20     92

Data Sorted by Marks:
    Name  Age  Marks
4   Neha   20     92
1  Sneha   21     90
0   Amit   20     89
3  Priya   22     88
2  Rahul   19     75

Average Marks:
86.8

Maximum Marks:
92

Minimum Marks:
75
```